Measure Theoretic Probability
Lesson 6
Extension of Probability Measures: Lemmas 1–3 on the Class of Measurable Sets

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Outline

1. Setting, Tools and Roadmap
2. Lemma 1: $\mathcal{M}$ Is a Field
3. Lemma 2: Additivity of the Outer Measure on $\mathcal{M}$
4. Lemma 3: $\mathcal{M}$ Is a Sigma-Field
5. Summary and Exercises

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The Setting and the Outer Measure

Setting

Ω is a non-empty set, $\mathcal{F}_0$ is a **field** of subsets of Ω, $\mathcal{F} = \sigma(\mathcal{F}_0)$, and $\mathbb{P}$ is a probability measure on $\mathcal{F}_0$.

Outer measure (defined on every subset of Ω)

$$\mathbb{P}^*(A) = \inf\Big\{\sum_{n=1}^{\infty} \mathbb{P}(A_n) : \ A_n \in \mathcal{F}_0, \ A \subseteq \bigcup_{n=1}^{\infty} A_n\Big\}.$$

Objective

Show that $\mathbb{P}^*$ restricted to $\mathcal{F}$ is a probability measure extending $\mathbb{P}$. The argument rests on six lemmas about a class $\mathcal{M}$ of subsets of Ω.

Available Tools: Properties of $\mathbb{P}^*$ Already Established

Established in Lesson 4

Label	Property	Statement (for arbitrary subsets)
(T1)	null set	$\mathbb{P}^*(\emptyset) = 0$
(T2)	monotonicity	$A \subseteq B \;\Rightarrow\; \mathbb{P}^*(A) \leq \mathbb{P}^*(B)$
(T3)	countable subadditivity	$\mathbb{P}^*\left(\bigcup_n A_n\right) \leq \sum_n \mathbb{P}^*(A_n)$
(T4)	bounded values	$0 \leq \mathbb{P}^*(A) \leq 1$

Consequence (T3′): finite subadditivity

Pad a finite family with $\emptyset, \emptyset, \ldots$ and use (T1) and (T3): $\mathbb{P}^*(A_1 \cup \cdots \cup A_n) \leq \sum_{k=1}^{n} \mathbb{P}^*(A_k)$.

Caution

We do **not** yet know that $\mathbb{P}^*$ is a probability measure. No additivity and no complement rule may be assumed: only (T1)–(T4).

The Class of Outer-Measurable Sets

Definition

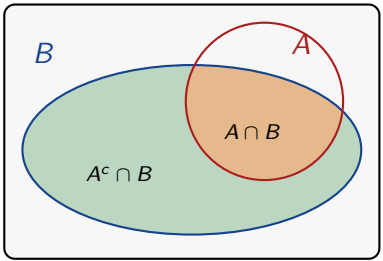
$$\mathcal{M} = \Big\{ A \subseteq \Omega : \ \mathbb{P}^*(A \cap B) + \mathbb{P}^*(A^c \cap B) = \mathbb{P}^*(B) \ \forall \, B \subseteq \Omega \Big\}.$$

Interpretation

A set $A \in \mathcal{M}$ **splits every test set** B into two parts whose outer measures add up to $\mathbb{P}^*(B)$. For a probability measure this identity is automatic; for $\mathbb{P}^*$ it is a genuine condition, and $\mathcal{M}$ is the class where it holds.

Terminology

Sets of $\mathcal{M}$ are called $\mathbb{P}^*$-measurable (Carathéodory-measurable) sets.



A One-Sided Criterion for Membership in $\mathcal{M}$

Proposition

$A \in \mathcal{M}$ if and only if for every $B \subseteq \Omega$,

$$\mathbb{P}^*(A \cap B) + \mathbb{P}^*(A^c \cap B) \;\leq\; \mathbb{P}^*(B).$$

Proof

For every A and B we have $B = (A \cap B) \cup (A^c \cap B)$. By finite subadditivity (T3′),

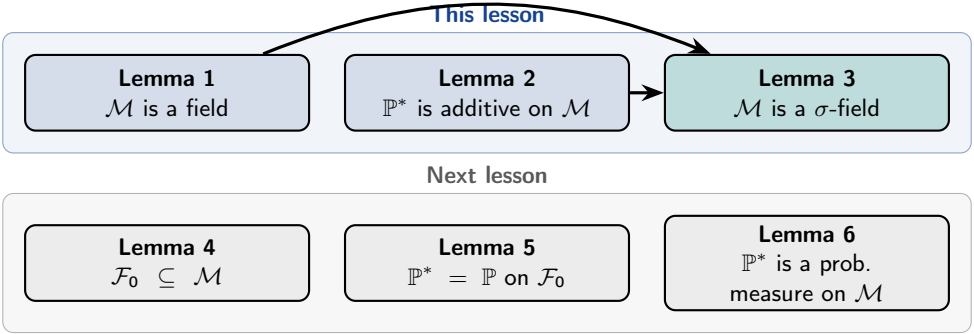
$$\mathbb{P}^*(B) \;\leq\; \mathbb{P}^*(A \cap B) + \mathbb{P}^*(A^c \cap B).$$

So the inequality "$\geq$" in the definition of $\mathcal{M}$ always holds, and only "$\leq$" remains to be checked. □

How the two forms are used

To use that $A \in \mathcal{M}$, we use the **equality**. To prove that $A \in \mathcal{M}$, it suffices to prove the **inequality**.

The Six Lemmas



Lemmas 1 and 2 together give Lemma 3: $\mathcal{M}$ is a σ-field.

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Lemma 1: Statement and Properties (i)–(ii)

Lemma 1

$\mathcal{M}$ is a field: (i) $\Omega \in \mathcal{M}$; (ii) $A \in \mathcal{M} \Rightarrow A^c \in \mathcal{M}$; (iii) $A_1, \ldots, A_n \in \mathcal{M} \Rightarrow \bigcup_{k=1}^{n} A_k \in \mathcal{M}$.

Proof of (i)

Let $B \subseteq \Omega$. Then $\Omega \cap B = B$ and $\Omega^c \cap B = \emptyset \cap B = \emptyset$, so by (T1)

$$\mathbb{P}^*(\Omega \cap B) + \mathbb{P}^*(\Omega^c \cap B) = \mathbb{P}^*(B) + \mathbb{P}^*(\emptyset) = \mathbb{P}^*(B).$$

Hence $\Omega \in \mathcal{M}$.

Proof of (ii)

Since $(A^c)^c = A$, the defining expression for A^c is $\mathbb{P}^*(A^c \cap B) + \mathbb{P}^*(A \cap B)$, the same expression as for A with the two terms swapped. So $A \in \mathcal{M} \Rightarrow A^c \in \mathcal{M}$.

Property (iii): Reduction to Intersections

Claim

It suffices to prove: $A_1, A_2 \in \mathcal{M} \;\Rightarrow\; A_1 \cap A_2 \in \mathcal{M}$.

Why this suffices

By De Morgan's law,

$$A_1 \cup A_2 = \left(A_1^c \cap A_2^c\right)^c.$$

If $A_1, A_2 \in \mathcal{M}$, then $A_1^c, A_2^c \in \mathcal{M}$ by (ii); their intersection is in $\mathcal{M}$ by the claim; and its complement is in $\mathcal{M}$ by (ii). So $A_1 \cup A_2 \in \mathcal{M}$.

From two sets to finitely many

Take unions two at a time: $A_1 \cup \cdots \cup A_n = \left(A_1 \cup \cdots \cup A_{n-1}\right) \cup A_n$. By induction on n, every finite union of sets of $\mathcal{M}$ lies in $\mathcal{M}$.

Intersections: Splitting the Test Set

Setup

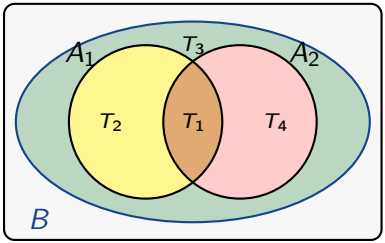
Let $A_1, A_2 \in \mathcal{M}$ and fix $B \subseteq \Omega$. By the criterion we must show

$$\mathbb{P}^*(A_1 \cap A_2 \cap B) + \mathbb{P}^*\big((A_1 \cap A_2)^c \cap B\big) \leq \mathbb{P}^*(B).$$

Four disjoint pieces of B

$T_1 = A_1 \cap A_2 \cap B$, $\quad T_2 = A_1 \cap A_2^c \cap B$,
$T_3 = A_1^c \cap A_2^c \cap B$, $\quad T_4 = A_1^c \cap A_2 \cap B$.

Then $(A_1 \cap A_2)^c \cap B = T_2 \cup T_3 \cup T_4$.



By subadditivity (T3′)

$\mathbb{P}^*(T_2 \cup T_3 \cup T_4) \leq \mathbb{P}^*(T_2) + \mathbb{P}^*(T_3) + \mathbb{P}^*(T_4)$, hence the left side above is $\leq \mathbb{P}^*(T_1) + \mathbb{P}^*(T_2) + \mathbb{P}^*(T_3) + \mathbb{P}^*(T_4)$.

Intersections: Grouping the Four Terms

Use $A_1 \in \mathcal{M}$ twice, with two different test sets

$$\mathbb{P}^*(T_1) + \mathbb{P}^*(T_4) = \mathbb{P}^*\big(A_1 \cap (A_2 \cap B)\big) + \mathbb{P}^*\big(A_1^c \cap (A_2 \cap B)\big) = \mathbb{P}^*(A_2 \cap B),$$
$$\mathbb{P}^*(T_2) + \mathbb{P}^*(T_3) = \mathbb{P}^*\big(A_1 \cap (A_2^c \cap B)\big) + \mathbb{P}^*\big(A_1^c \cap (A_2^c \cap B)\big) = \mathbb{P}^*(A_2^c \cap B).$$

Then use $A_2 \in \mathcal{M}$ with test set B

$$\sum_{i=1}^{4} \mathbb{P}^*(T_i) = \mathbb{P}^*(A_2 \cap B) + \mathbb{P}^*(A_2^c \cap B) = \mathbb{P}^*(B).$$

Conclusion

$\mathbb{P}^*(A_1 \cap A_2 \cap B) + \mathbb{P}^*\big((A_1 \cap A_2)^c \cap B\big) \leq \mathbb{P}^*(B)$ for every B, so $A_1 \cap A_2 \in \mathcal{M}$. With the reduction, (iii) holds and **Lemma 1 is proved**. □

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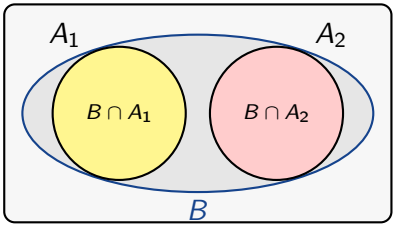
Lemma 2: Statement

Lemma 2

Let $A_1, A_2, \cdots \in \mathcal{M}$ be **pairwise disjoint** and let $B \subseteq \Omega$ be arbitrary. Then

(a) $\mathbb{P}^*\Big(B \cap \bigcup_{k=1}^{n} A_k\Big) = \sum_{k=1}^{n} \mathbb{P}^*(B \cap A_k)$ for every n;

(b) $\mathbb{P}^*\Big(B \cap \bigcup_{k=1}^{\infty} A_k\Big) = \sum_{k=1}^{\infty} \mathbb{P}^*(B \cap A_k)$.



Special case $B = \Omega$

$\mathbb{P}^*\big(\bigcup_k A_k\big) = \sum_k \mathbb{P}^*(A_k)$: additivity of $\mathbb{P}^*$ on disjoint sets of $\mathcal{M}$. The stronger form with a test set B is needed in Lemma 3.

Proof of (a): Induction on n

Base case $n = 1$

Both sides equal $\mathbb{P}^*(B \cap A_1)$.

Induction step

Assume (a) holds for $n-1$ and let $C = \bigcup_{k=1}^{n} A_k$. Since $A_n \in \mathcal{M}$, apply its defining equality with the test set $B \cap C$:

$$\mathbb{P}^*(B \cap C) = \mathbb{P}^*(B \cap C \cap A_n) + \mathbb{P}^*(B \cap C \cap A_n^c).$$

By disjointness, $C \cap A_n = A_n$ and $C \cap A_n^c = \bigcup_{k=1}^{n-1} A_k$. Therefore, by the induction hypothesis,

$$\mathbb{P}^*(B \cap C) = \mathbb{P}^*(B \cap A_n) + \mathbb{P}^*\Big(B \cap \bigcup_{k=1}^{n-1} A_k\Big) = \sum_{k=1}^{n} \mathbb{P}^*(B \cap A_k). \qquad \square$$

Proof of (b): Two Inequalities

($\leq$) by subadditivity

Since $B \cap \bigcup_k A_k = \bigcup_k (B \cap A_k)$, (T3) gives

$$\mathbb{P}^*\Big(B \cap \bigcup_k A_k\Big) \leq \sum_{k=1}^{\infty} \mathbb{P}^*(B \cap A_k).$$

($\geq$) by monotonicity

For each N, $B \cap \bigcup_{k \leq N} A_k \subseteq B \cap \bigcup_k A_k$, so by (T2) and part (a),

$$\mathbb{P}^*\Big(B \cap \bigcup_k A_k\Big) \geq \sum_{k=1}^{N} \mathbb{P}^*(B \cap A_k).$$

Passage to the limit

The left side does not depend on N. Letting $N \to \infty$ turns the partial sums into the infinite series, so $\mathbb{P}^*\big(B \cap \bigcup_k A_k\big) \geq \sum_{k=1}^{\infty} \mathbb{P}^*(B \cap A_k)$. Together with ($\leq$), equality holds. $\square$

Remark

Only the definition of $\mathcal{M}$ and (T1)–(T3) were used: no property of probability measures was assumed for $\mathbb{P}^*$.

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Lemma 3: Statement and Reduction to Disjoint Sets

Lemma 3

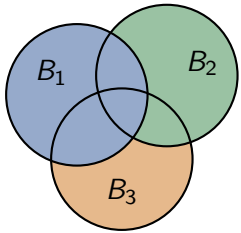
$\mathcal{M}$ is a σ-field. By Lemma 1, $\Omega \in \mathcal{M}$ and $\mathcal{M}$ is closed under complements, so it remains to show closure under **countable** unions.

Reduction to disjoint sets

Let $A_1, A_2, \cdots \in \mathcal{M}$. Put $B_1 = A_1$ and, for $n \geq 2$,

$$B_n = A_n \cap A_1^c \cap \cdots \cap A_{n-1}^c.$$

Then $B_n \in \mathcal{M}$ (Lemma 1), the B_n are pairwise disjoint, and $\bigcup_n B_n = \bigcup_n A_n$.



Blue: A_1. Green: $A_2 \setminus A_1$.
Orange: $A_3 \setminus (A_1 \cup A_2)$.

Consequence

It is enough to prove the claim for **pairwise disjoint** $A_n \in \mathcal{M}$.

Proof of Lemma 3: The Key Inequality

Setup

Let $A_1, A_2, \cdots \in \mathcal{M}$ be pairwise disjoint, $A = \bigcup_{n=1}^{\infty} A_n$, and $C_m = \bigcup_{n=1}^{m} A_n$. Then $C_m \in \mathcal{M}$ by Lemma 1. Fix $B \subseteq \Omega$.

Three facts

(1) $\mathbb{P}^*(B) = \mathbb{P}^*(C_m \cap B) + \mathbb{P}^*(C_m^c \cap B)$, since $C_m \in \mathcal{M}$.

(2) $\mathbb{P}^*(C_m \cap B) = \sum_{n=1}^{m} \mathbb{P}^*(A_n \cap B)$, by Lemma 2(a).

(3) $A^c \subseteq C_m^c$, so $\mathbb{P}^*(C_m^c \cap B) \geq \mathbb{P}^*(A^c \cap B)$, by (T2).

Combining (1)–(3)

$$\mathbb{P}^*(B) \;\geq\; \sum_{n=1}^{m} \mathbb{P}^*(A_n \cap B) \;+\; \mathbb{P}^*(A^c \cap B) \qquad \text{for every } m.$$

Proof of Lemma 3: Conclusion

Let $m \to \infty$

The left side does not depend on m, and the partial sums converge to the infinite series, so

$$\mathbb{P}^*(B) \;\geq\; \sum_{n=1}^{\infty} \mathbb{P}^*(A_n \cap B) \;+\; \mathbb{P}^*(A^c \cap B) \;=\; \mathbb{P}^*(A \cap B) + \mathbb{P}^*(A^c \cap B),$$

where the last equality is Lemma 2(b) applied to the disjoint sets $A_n \in \mathcal{M}$ and the test set B.

Conclusion

The inequality holds for every $B \subseteq \Omega$, so $A = \bigcup_n A_n \in \mathcal{M}$ by the one-sided criterion. Thus $\mathcal{M}$ is closed under countable unions and, with Lemma 1, $\mathcal{M}$ **is a σ-field**. □

Why Lemma 2 carries a test set B

We cannot assume that $A_n \cap B$ lies in $\mathcal{M}$, so we cannot treat these sets as members of $\mathcal{M}$. Lemma 2(b) gives the needed additivity for arbitrary B. (Subadditivity (T3) would also give the final inequality.)

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Summary of Lemmas 1–3

Key take aways

Lemma	Statement	what goes into them
1	$\mathcal{M}$ is a field	(T1), De Morgan, subadditivity, regrouping of four pieces
2	$\mathbb{P}^*(B \cap \bigcup A_k) = \sum \mathbb{P}^*(B \cap A_k)$ for disjoint $A_k \in \mathcal{M}$	induction; monotonicity and limits of partial sums
3	$\mathcal{M}$ is a σ-field	disjointification

Result of this lesson

$\mathbb{P}^*$ is **countably additive** on the σ-field $\mathcal{M}$ (Lemma 2 with $B = \Omega$, Lemma 3).

Exercises

1. Show that every $A \subseteq \Omega$ with $\mathbb{P}^*(A) = 0$ belongs to $\mathcal{M}$.
2. Write out the induction proving that $A_1, \ldots, A_n \in \mathcal{M} \Rightarrow A_1 \cup \cdots \cup A_n \in \mathcal{M}$.
3. Show that $A, B \in \mathcal{M} \Rightarrow A \setminus B \in \mathcal{M}$.
4. Show that $\mathbb{P}^*(A \cup B) = \mathbb{P}^*(A) + \mathbb{P}^*(B)$ whenever $A, B \in \mathcal{M}$ are disjoint.

Outlook

Next lesson: Lemmas 4–6

- **Lemma 4:** $\mathcal{F}_0 \subseteq \mathcal{M}$.
- **Lemma 5:** $\mathbb{P}^*(A) = \mathbb{P}(A)$ for $A \in \mathcal{F}_0$
- **Lemma 6:** $\mathbb{P}^*$ is a probability measure on $(\Omega, \mathcal{M})$.

How the pieces fit

Lemmas 3 and 4 give $\mathcal{F}_0 \subseteq \mathcal{F} = \sigma(\mathcal{F}_0) \subseteq \mathcal{M}$ (the generated σ-field is the smallest one).
Then $\mathbb{P}^*$ restricted to $\mathcal{F}$ is a probability measure extending $\mathbb{P}$.